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APA – Real Time Applications

Module 4 – Chat Bot Techniques using Scikit Learn

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Agenda – Module: Introduction to ChatBot & its Application

1. Overview of Chatbot Applications
2. Natural Language Processing for Chatbots
3. Building Chatbots the Easy Way
4. Building the Chatbot the Hard way
5. Deploying Your Chatbot using Python Applications
6. Learning Model Evaluation and Improvement Process

Introduction to ChatBot & its Application

Module Objective:

- Understand the role and benefits of chatbots in Real World Applications esp. in digital marketing.
- Learn about different types of chatbots and their applications.
- Explore practical examples and use cases.
- Gain hands-on experience in creating and deploying a basic Chatbot.

Overview of ChatBot

This Module is for learning the concepts related to chatbots and learning how to build them

Bots in general are nothing but a machine that is intelligent enough to understand your request and then formulate your request in such a way that is understandable by other software systems to request the data you need.

Reason Chatbots get an edge over traditional methods of getting things done online is you can do multiple things with the help of a chatbot. It's not just a chatbot, it's like your virtual personal assistant. You can think of being able to book a hotel room on booking.com as well as booking a table in a nearby restaurant of the hotel, but you can do that using your chatbot. Chatbots fulfill the need of being multipurpose and hence save a lot of time and money.

<u>Two of the most widely adopted machine learning methods are</u>

- **Supervised learning** are trained using labeled examples, such as an input where the desired output is known e.g. regression or classification
- **Unsupervised learning** is used against data that has no historical labels e.g. cluster analysis
- Third ML paradigm **is Semi-supervised learning** which is used when there are strong reasons to believe that a typical pattern exists in data such that the given pattern can be quantified via models.

Introduction to Chatbot

Objective: Understand the fundamentals of chatbots and NLP, and their applications in digital marketing.

What Are Chatbots?

- **Definition:** AI-powered software designed to simulate conversation with users.

Types of Chatbots:

- **Rule-Based:** Follow pre-set rules and scripts.
- **AI-Powered:** Utilize natural language processing (NLP) and machine learning.

Key Characteristics:

- **24/7 Availability**
- **Instant Responses**
- **Automated Interaction**

Evolution of Chatbots

Historical Development:

- ✓ **Early Chatbots:** Simple scripts and decision trees.
- ✓ **Advancements:** NLP, machine learning, and integration with other AI technologies.
- ✓ **Current Trends:** Personalization, contextual understanding, and multi-channel integration.

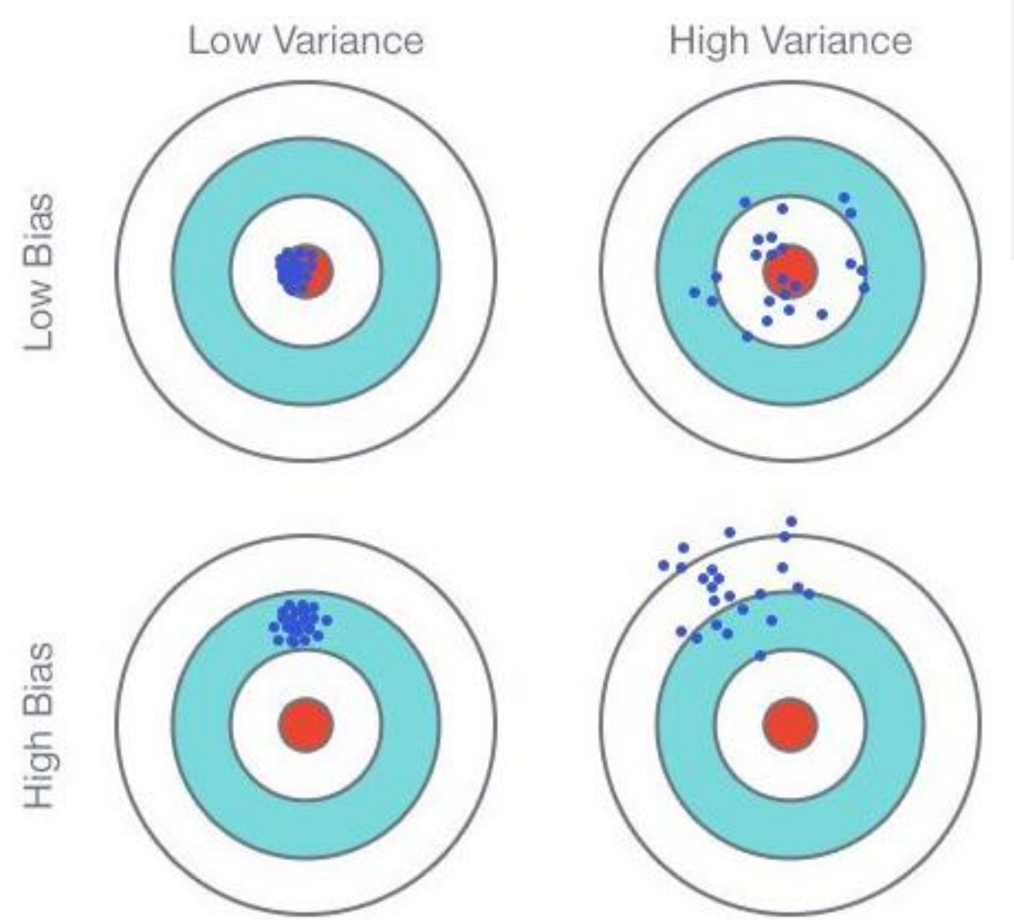
Impact on User Experience:

- **Enhanced Engagement**
- **Increased Efficiency**

Examples of ChatBot

- Low-bias algorithms include: Decision Trees, k-Nearest Neighbors, Support Vector Machines
- High-bias algorithms include: Linear Regression, Linear Discriminant Analysis, Logistic Regression

- **Bias** can be considered as **Accuracy**
- **Variance** can be considered as **Stability**



Role of Chatbots in Digital Analytics World

1. Customer Support:

- **Instant Assistance:** Address queries and resolve issues quickly.
- **Consistent Service:** Maintain uniform responses and availability.

2. Lead Generation:

- **Qualification:** Engage visitors and gather information.
- **Conversion:** Direct users to relevant products or services.

3. Personalization:

- **Tailored Interactions:** Offer recommendations based on user behavior.
- **Customized Offers:** Provide personalized deals and promotions.

4. Data Collection:

- **Insights:** Gather valuable data on customer preferences and behavior.
- **Analytics:** Use data to refine marketing strategies.

Benefits of Chatbots

1. Cost Efficiency:

- ***Reduced Operational Costs:*** Automate repetitive tasks and reduce the need for a large customer support team.

2. Enhanced User Experience:

- ***Quick Resolution:*** Speed up response times and increase user satisfaction.
- ***Seamless Interaction:*** Provide a smooth and engaging experience.

3. Scalability:

- ***Handle High Volume:*** Manage multiple interactions simultaneously without additional resources.

4. Increased Engagement:

- ***Proactive Outreach:*** Initiate conversations and drive user interaction.

Examples of Successful Chatbot Implementations

E-Commerce:

Example: Sephora's chatbot for product recommendations and booking appointments.

Travel:

Example: KLM's chatbot for booking confirmations and flight information.

Banking:

Example: Bank of America's chatbot for account management and customer support.

Key Considerations for Implementing Chatbots

1. Define Objectives:

- **Purpose:** Identify what you want to achieve (e.g., support, lead generation).

2. Choose the Right Technology:

- **Platform:** Select based on functionality and integration capabilities.

3. Design for User Experience:

- **Conversation Flow:** Ensure interactions are intuitive and user-friendly.

4. Monitor and Optimize:

- **Performance Tracking:** Analyze metrics and continuously improve the chatbot’s performance.

Future Trends in Chatbots

Chatbots are transforming digital marketing by enhancing customer engagement, streamlining processes, and providing valuable insights.

Future Outlook: Continuous advancements will further expand their capabilities and applications. Couple of them are listed below:

Advanced AI Capabilities:

Natural Language Understanding (NLU): More accurate and nuanced conversations.

Integration with Emerging Technologies:

Voice Assistants: Integration with voice-controlled devices.

Greater Personalization:

Contextual Awareness: Enhanced ability to understand and anticipate user needs.

Introduction to Natural Language Processing (NLP)

What is NLP?

- **Definition:** A branch of artificial intelligence that enables computers to understand, interpret, and generate human language.

Core Components:

- **Text Analysis**
- **Speech Recognition**
- **Language Generation**

Importance in Chatbots:

- **Enhanced Understanding:** Comprehend user queries more accurately.
- **Improved Responses:** Generate contextually relevant replies.
- **Personalization:** Tailor interactions based on user language and intent.

How NLP Enhances Chatbot Capabilities

1. **Contextual Understanding:**
   - **Intent Recognition:** Determine user goals and needs from their input.
   - **Entity Extraction:** Identify key information (e.g., dates, locations).
2. **Sentiment Analysis:**
   - **Emotional Tone:** Assess user emotions (positive, negative, neutral).
   - **Response Adjustment:** Adapt chatbot responses based on sentiment.
3. **Language Generation:**
   - **Natural Responses:** Craft replies that mimic human-like conversational patterns.
   - **Dynamic Interaction:** Adjust language style based on user profile.

Understand Chatbot functionality via Live Case

Explaining via Video – Chatbot

https://www.youtube.com/watch?v=o9-ObGgfpEk

Case Study – breakout session for 20mins

Case Study: Customer Insights and Personalization

- **Company:** [Example Company]
  - **Use Case:** Analyzing chatbot interactions to refine marketing strategies.
  - **Approach:** Used NLP to identify customer preferences and behavior patterns.
  - **Outcome:** Increased engagement by offering personalized product recommendations and targeted promotions.

Key Considerations for Implementing NLP in Chatbots

- **Data Quality:**
  - **Accuracy:** Ensure high-quality training data for effective NLP performance.
- **Integration:**
  - **Systems:** Seamlessly integrate NLP capabilities with existing chatbot platforms and marketing tools.
- **Privacy and Compliance:**
  - **Data Handling:** Follow regulations for handling user data and privacy.
- **Continuous Improvement:**
  - **Monitoring:** Regularly update NLP models based on user feedback and evolving language trends.

Chatbots enhance digital marketing through automation and engagement; NLP further refines these capabilities by improving understanding and personalization.

Future Outlook: Continued advancements in NLP will drive more sophisticated and effective chatbot interactions.

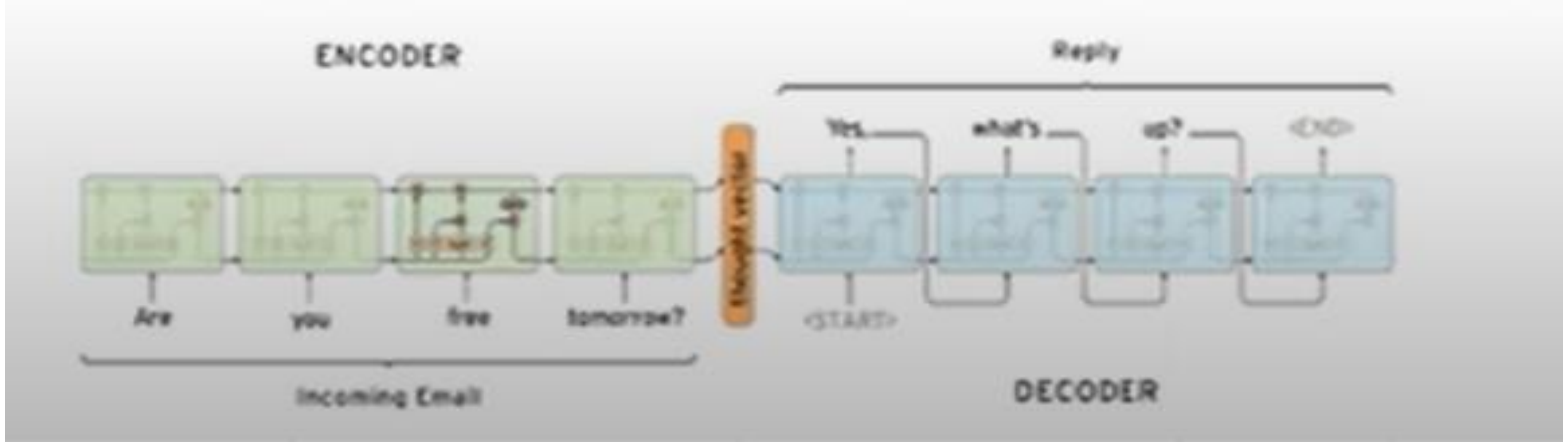
Types of Chatbot Models

RETRIEVAL-BASED VS. GENERATIVE MODELS

Retrieval-based models (easier) use a repository of predefined responses and some kind of heuristic to pick an appropriate response based on the input and context.

The heuristic could be as simple as a rule-based expression match, or as complex as an ensemble of Machine Learning classifiers.. These systems don't generate any new text, they just pick a response from a fixed set.

Generative models (harder) don't rely on pre-defined responses. They generate new responses from scratch… Generative models are typically based on Machine Translation techniques, but instead of translating from one language to another, we "translate" from an input to an output (response).



Retrieval-based vs. Generative Models – Pros and Cons

Pros and cons:

- Due to the repository of handcrafted responses, retrieval-based methods don't make grammatical mistakes. However, they may be unable to handle unseen cases for which no appropriate predefined response exists. For the same reasons, these models can’t refer back to contextual entity information like names mentioned earlier in the conversation.
- Generative models are "smarter". They can refer back to entities in the input and give the impression that you're talking to a human. However, these models are hard to train, are quite likely to make grammatical mistakes (especially on longer sentences), and typically require huge amounts of training data.

Deep Learning techniques can be used for both retrieval-based or generative models, but research seems to be moving into the generative direction. Production systems are more likely to be retrieval-based for now.

Long vs. Short Conversations in Chatbot

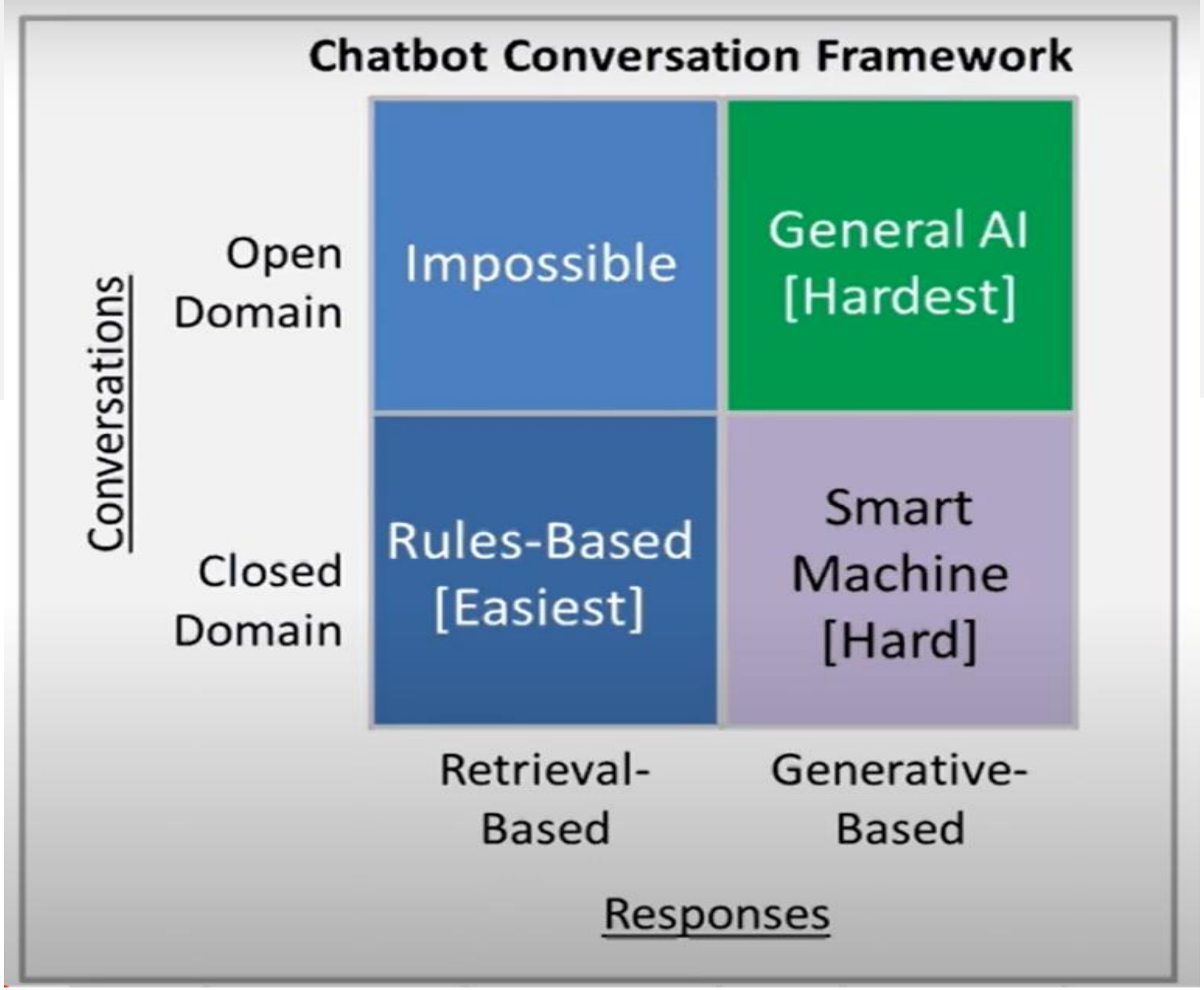
Long Conversation : More difficult to automate it.

Short Conversation: The goal is to create a single response to a single input. Easy to manage.

Short conversations are usually single turn.

Then there are **long** conversations (harder) where you go through multiple turns and need to keep track of what has been said. Customer support conversations are typically long conversational threads with multiple questions.

Chatbots: Open Domain vs. Closed Domain



Open vs. Closed Domain

Open domain

- In an **open domain (harder)** setting the user can take the conversation anywhere.
- There isn't necessarily have a well-defined goal or intention.
- Conversations on social media sites like Twitter and Reddit are typically open domain – they can go into all kinds of directions.
- The infinite number of topics and the fact that a certain amount of world knowledge is required to create reasonable responses makes this a hard problem.

Closed domain.

- In a **closed domain (easier)** setting the space of possible inputs and outputs is somewhat limited because the system is trying to achieve a very specific goal.
- Technical Customer Support or Shopping Assistants are examples of closed domain problems.
- These systems don't need to be able to talk about politics, they just need to fulfill their specific task as efficiently as possible.

Chatbots: Common Challenges – 1. Incorporating Context

- To produce sensible responses systems = Need to incorporate both ***linguistic context*** and ***physical context.***
- In long dialogs people keep track of what has been said and what information has been exchanged. That's an example of ***linguistic context***.
- One may also need to incorporate other kinds of contextual data such as ***date/time, location, or information*** about a user.
- Longer the conversations and the more important the context, the more difficult the problem becomes.

Chatbots: Common Challenges – 2. Coherent Personality

Chatbot can be giving completely inconsistent answers for same or similar set of questions – as mentioned below:

message	Where do you live now?
response	I live in Los Angeles.
message	In which city do you live now?
response	I live in Madrid.
message	In which country do you live now?
response	England, you?

Tip: When generating responses the agent should ideally produce consistent answers to semantically identical inputs.

Chatbots: Common Challenges – 3. Evaluation of Models

- Best **Measurement** process : is to measure whether or not it is fulfilling its task.
  - **Example** : Solve a customer support problem.

- But such labels are expensive to obtain because they require human judgment and evaluation. Sometimes there is no well-defined goal, as is the case with open-domain models.

- Common metrics such as BLEU* that are used for Machine Translation and are based on text matching aren't well suited because sensible responses can contain completely different words or phrases.

***- BLEU** stands for **Bilingual Evaluation Understudy (refer appendix for details)**. It is a widely used automatic metric for evaluating the quality of text that has been machine-translated or generated by natural language processing (NLP) models.

- For chatbots, BLEU can help assess how closely the generated responses match human-like reference responses, though it's often used in conjunction with other metrics and human evaluations for a more comprehensive assessment.

Chatbots: Common Challenges – 4. Intension and Diversity

- Problem with generative systems is that they tend to produce generic responses like "That's great!" or "I don't know" that work for a lot of input cases.
- Generative systems aren't trained to have specific intentions they lack this kind of diversity.
- However, humans typically produce responses that are specific to the input and carry an intention. Because generative systems (and particularly open-domain systems) aren't trained to have specific intentions they lack this kind of diversity.

Current Status of Chatbot

- Many companies ***start off by outsourcing their conversations to human workers*** and promise that they can "automate" it once they've collected enough data.
- That's likely to happen only if they are operating in a pretty narrow domain – like a chat interface to call an Uber for example.
- Anything that's a ***bit more open domain (like sales emails)*** is beyond what we can currently do. However, we can also ***use these systems to assist human workers by proposing and correcting responses***. That's much more feasible.
- Grammatical mistakes in production systems are very costly and may drive away users. That's why ***most systems are probably best-off using retrieval-based methods*** that are ***free of grammatical errors and offensive responses***.
- If companies can somehow get their hands on huge amounts of data then generative models become feasible - but they must be assisted by other techniques to prevent them from going off the rails like Microsoft's Tay did.

Thank You

Appendix

What is BLEU (Bilingual Evaluation Understudy) score

The BLEU (Bilingual Evaluation Understudy) score is a metric used to evaluate the quality of text generated by machine translation systems and, by extension, chatbots that generate text. Here's a quick overview of what it involves:

What BLEU Measures:

- **Precision:** BLEU measures how many of the n-grams (contiguous sequences of n words) in the machine-generated text appear in the reference texts (usually human-generated translations).
- **n-gram Matching:** It checks for matches of unigrams (1-grams), bigrams (2-grams), trigrams (3-grams), and sometimes higher-order n-grams.
- **Brevity Penalty:** To prevent systems from generating overly short translations that might artificially inflate the precision score, BLEU includes a brevity penalty. This penalizes translations that are shorter than the reference translations.

How BLEU is Calculated:

1. **Count Matches:** Count the number of overlapping n-grams between the candidate translation and the reference translations.
2. **Calculate Precision:** Compute precision for each n-gram level (e.g., unigram, bigram) by dividing the count of matching n-grams by the total number of n-grams in the candidate translation.
3. **Apply Brevity Penalty:** Adjust the precision score by a brevity penalty if the candidate translation is shorter than the reference translations.
4. **Combine Scores:** BLEU combines these precision scores using a geometric mean, weighted by the importance of each n-gram level.

Limitations of BLEU:

- **Lacks Semantic Understanding:** BLEU focuses on exact n-gram matching and doesn’t account for synonyms, paraphrases, or the overall meaning.
- **Doesn’t Capture Fluency:** It doesn't assess the fluency or grammatical correctness of the generated text.
- **Reference Sensitivity:** The score can be sensitive to the choice of reference translations and might not reflect the quality if references are too different.

Despite these limitations, BLEU remains a widely used and accepted metric for evaluating machine translation and text generation systems due to its simplicity and ease of computation. For chatbots, BLEU can help assess how closely the generated responses match human-like reference responses, though it’s often used in conjunction with other metrics and human evaluations for a more comprehensive assessment.

Confusion matrix

Definition: A Confusion Matrix is a table used to evaluate the performance of a classification model. It summarizes the prediction results by comparing the predicted labels with the true labels. It provides insights into how many instances were correctly or incorrectly classified by the model.

Components:

- **True Positive (TP):** Number of instances correctly predicted as positive.
- **True Negative (TN):** Number of instances correctly predicted as negative.
- **False Positive (FP):** Number of instances incorrectly predicted as positive (Type I error).
- **False Negative (FN):** Number of instances incorrectly predicted as negative (Type II error).

Purpose: To assess the performance of a classification model by providing metrics like accuracy, precision, recall, and F1-score.

References

- https://www.youtube.com/watch?v=ci2EOOxyKQo